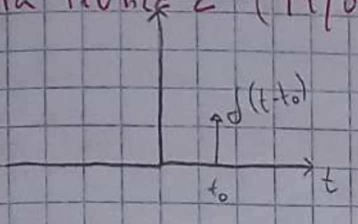
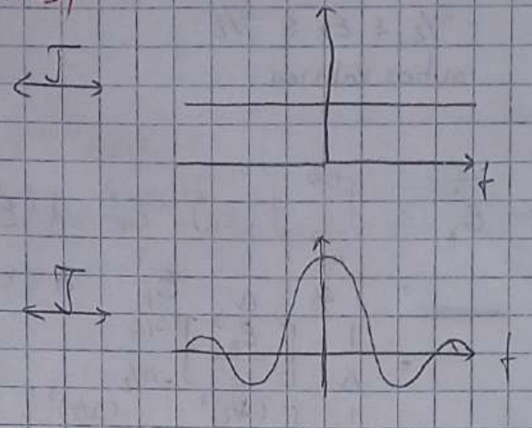
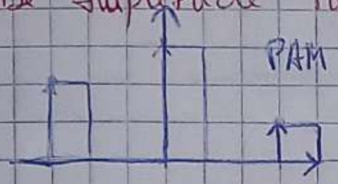
Aula Teórica 1 (12/09/2023)

Aula Teórica 2 (19/09/2023)





Pulse Amplitude Modulation



Sample - and - hold

Se é só impulso → tem largura de banda infinita

Se é impulso limitado no tempo → tem largura de banda "finita"

O PAM não é ~~uma~~ usado como modulação final, mas como uma modulação intermédia, utilizando-se ainda outra modulação antes de enviar o sinal

O PPM é melhor que o PDM porque gasta menos energia, uma vez que a largura dos impulsos é menor no PPM do que no PDM

Quantization process

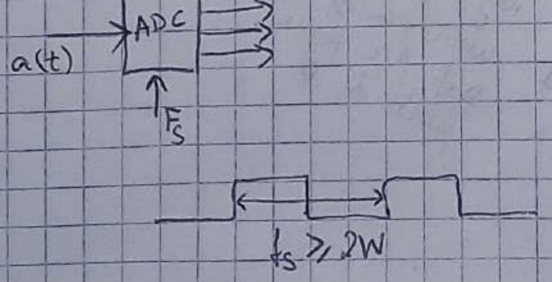
$q = 8$ (nº de níveis de quantização)

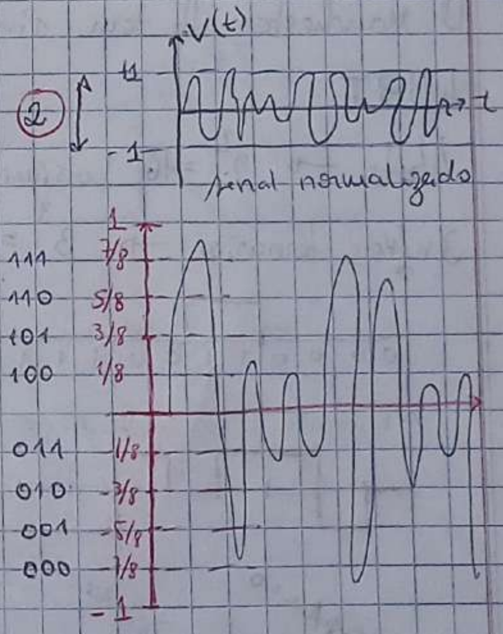
$\Delta = \frac{2}{q} = \frac{2}{8} = 0{,}25V$ (separação entre níveis)

níveis discretos são:

$\pm 1/8$; $\pm 3/8$; $\pm 5/8$; $\pm 7/8$

para $q = 8 \rightarrow 3$ bits





Erro de Quantização: $\varepsilon_K = x_q(kT_s) - x(kT_s)$

$$f(\varepsilon_K) = \begin{cases} 1/\Delta & -\Delta/2 \le \varepsilon_K \le \Delta/2 \\ 0 & \text{outros valores} \end{cases}$$

Potência $\Rightarrow$

$$\sigma_q^2 = \overline{\varepsilon_K^2} = \int_{-\infty}^{+\infty} f(\varepsilon_K) \cdot \varepsilon_K^2 \, d(\varepsilon_K)$$
$$= \int_{-\Delta/2}^{\Delta/2} \frac{1}{\Delta} \cdot \varepsilon_K^2 \, d(\varepsilon_K)$$
$$= \frac{1}{\Delta} \left[\frac{\varepsilon_K^3}{3} \right]_{-\Delta/2}^{\Delta/2}$$
$$= \frac{1}{\Delta} \left[\frac{(\Delta/2)^3}{3} + \frac{(\Delta/2)^3}{3} \right]$$
$$= \frac{1}{\Delta} \left[\frac{\Delta^3}{24} + \frac{\Delta^3}{24} \right]$$
$$= \frac{1}{\cancel{\Delta}} \left[\frac{\Delta^{\cancel{3}}}{12} \right]$$
$$= \frac{\Delta^2}{12}$$

$SNR = S_x / \sigma_q^2 \le 4{,}8 + 6V$ (Signal-to-noise Ratio)

Pulse Code Modulation (PCM)

Time Division Multiplexing (TDM)

Line Codes

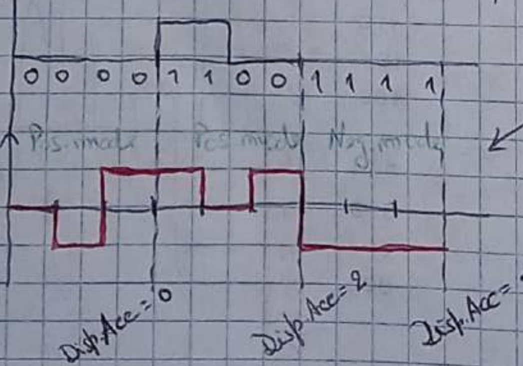
O CMI tem mais transições o que favorece a deteção do clock na receção

O Manchester tb tem ainda mais transições

4B3T

4bits $\rightarrow 2^4 = 16$ possíveis combinações binárias

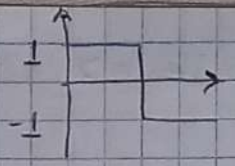
3 níveis ternários $\rightarrow 3^3 = 27$ possíveis combinações ternárias



tentar alternar entre ~~[illegible]~~ neg. e pos. de maneira q a Disp. Acc fique perto de 0

Delta Modulation

Sgn (t) → Função Sinal



Aula Teórica 3 (26/09/2023)

Matched Filter

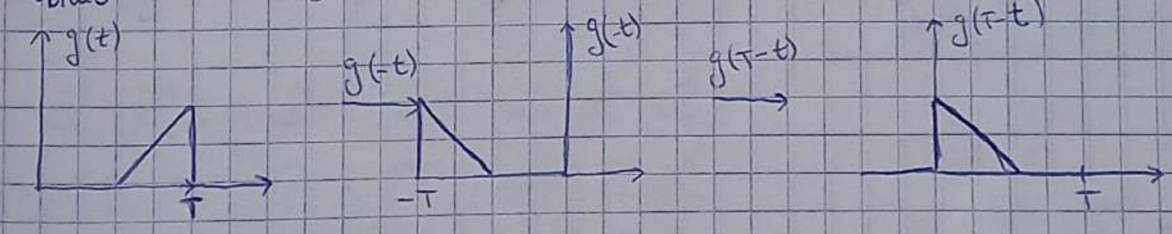
$G(f)$ → T. Fourier de $g(t)$

$H(f)$ → Resposta em frequência do Filtro

$G(f) \cdot H(f)$ → T. Fourier de $g_0(t)$

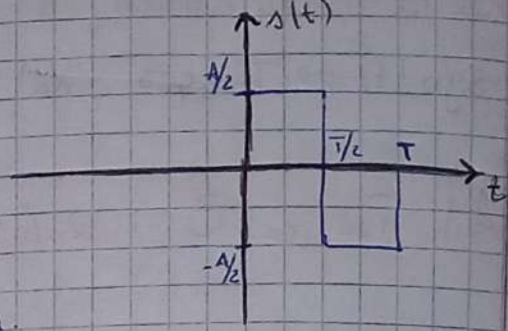
$$g_0(t) = \int_{-\infty}^{+\infty} H(f) \cdot G(f) \cdot e^{j2\pi f t} \, df.$$

$h_{ótimo}(t) = K \cdot g(T-t)$ (exemplo)



Exercício: <Pode sair no teste>

Considere o sinal $s(t)$:

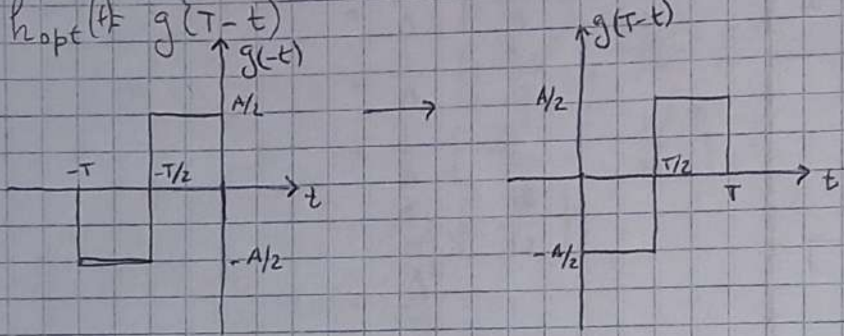


a) $h_{opt}(t) = ?$

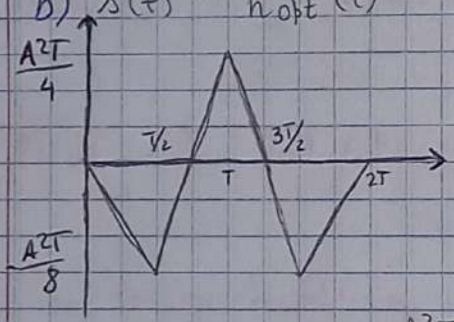
b) Desenhe a saída do Matched Filter

c) Qual o valor do pico de saída.

a) $h_{opt}(t) = g(T-t)$



b) $s(t) * h_{opt}(t)$



c) valor de pico $= \frac{A^2T}{4}$

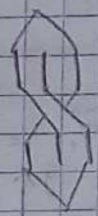
TPC provar que o valor da probabilidade de erro dos repetidores não-regenerativos é maior que os repetidores regenerativos

Aula Teórica 4 (3/10/2023)

$\vec{v}_1 = (1, 0, 1) \Rightarrow \|\vec{v}_1\| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$

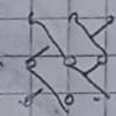
$\vec{v}_2 = (1, 1, 0) \Rightarrow \|\vec{v}_2\| = \sqrt{2}$

$\vec{v}_3 = (0, 1, 1) \Rightarrow \|\vec{v}_3\| = \sqrt{2}$



a) Verifica se são linearmente independentes?

b) Verifica se são ortogonais?



a) $a \cdot \vec{v}_1 + b \cdot \vec{v}_2 + c \cdot \vec{v}_3 = (0, 0, 0) \quad \wedge \quad a = b = c = 0$

$(a, 0, a) + (b, b, 0) + (0, c, c) = (0, 0, 0)$

$$\begin{cases} a + b = 0 \\ b + c = 0 \\ a + c = 0 \end{cases} \Leftrightarrow \begin{cases} a = -b \\ c = -b \\ -b - b = 0 \end{cases} \quad \begin{cases} a = 0 \\ c = 0 \\ b = 0 \end{cases}$$

Como são linearm// independentes, constituem uma base vetorial, $\mathbb{R}^3$, e qualquer vetor pode ser construido através de combinações lineares destes 3 vetores

b) $\vec{v}_1 \cdot \vec{v}_2 = \langle \vec{v}_1 \cdot \vec{v}_2 \rangle = \|\vec{v}_1\| \cdot \|\vec{v}_2\| \cdot \cos(\vec{v}_1, \vec{v}_2)$

$= (1, 0, 1) \cdot (1, 1, 0)$

$= 1 + 0 + 0 = 1$ $\bar{n}$ são ortogonais

$\vec{v}_2 \cdot \vec{v}_3 = (1, 1, 0) \cdot (0, 1, 1) = 0 + 1 + 0 = 1$, $\bar{n}$ são ortogonais

$\vec{v}_1 \cdot \vec{v}_3 = (1, 0, 1) \cdot (0, 1, 1) = 0 + 0 + 1 = 1$, $\bar{n}$ são ortogonais

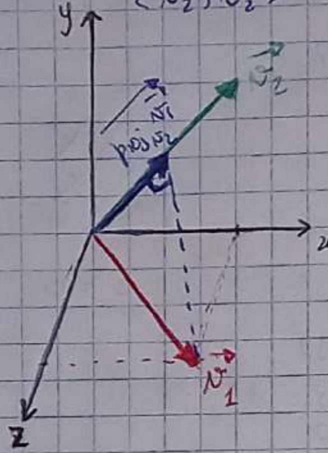
$\vec{v}_1 \cdot \vec{v}_2 = \|\vec{v}_1\| \cdot \|\vec{v}_2\| \cdot \cos(\vec{v}_1, \vec{v}_2)$

$\Leftrightarrow 1 = \sqrt{2} \cdot \sqrt{2} \cdot \cos(\vec{v}_1, \vec{v}_2)$

$\Leftrightarrow \cos(\vec{v}_1, \vec{v}_2) = \frac{1}{2} \Leftrightarrow \measuredangle(\vec{v}_1, \vec{v}_2) = 60°$

$\text{projeção}_{\vec{u}} \vec{v} = \frac{\langle \vec{v}, \vec{u} \rangle}{\langle \vec{u}, \vec{u} \rangle} \cdot \vec{u} \Rightarrow \vec{x}$

$\text{proj}_{\vec{v}_2} \vec{v}_1 = \frac{\langle \vec{v}_1 , \vec{v}_2 \rangle}{\langle \vec{v}_2 , \vec{v}_2 \rangle} \vec{v}_2 = \frac{1}{2} \cdot (1,1,0) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$

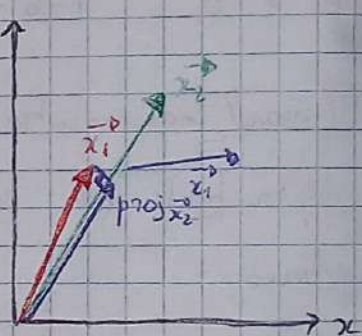


$\vec{x}_1 = (1,2) \qquad \vec{x}_2 = (2,3)$

$\text{proj}_{\vec{x}_2} \vec{x}_1 = \frac{\langle x_1, x_2 \rangle}{\langle x_2, x_2 \rangle} \cdot \vec{x}_2$

$\vec{a}_{\vec{x}_1} = \frac{8}{13} (2,3)$

$\vec{a} = \text{proj}_{\vec{x}_2} = \left(\frac{16}{13}, \frac{24}{13}\right)$



Gram-Schmidt Ortogonalization

$\vec{u}_1 = \vec{v}_1 = (1,0,1)$

$\vec{u}_2 = \vec{v}_2 - \text{proj}_{\vec{u}_1}(\vec{v}_2) = (1,1,0) - \frac{(1,1,0)\cdot(1,0,1)}{(1,0,1)\cdot(1,0,1)} \cdot (1,0,1)$

$= (1,1,0) - \frac{1}{2} \cdot (1,0,1)$

$= \left(\frac{1}{2}, 1, -\frac{1}{2}\right)$

$\vec{u}_3 = \vec{v}_3 - \text{proj}_{\vec{u}_1}(\vec{v}_3) - \text{proj}_{\vec{u}_2}(\vec{v}_3) = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$

$\|\vec{u}_1\| = \sqrt{2} \Rightarrow \vec{x}_1 = \frac{\vec{u}_1}{\|\vec{u}_1\|} = \frac{(1,0,1)}{\sqrt{2}} = \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right)$

$\|\vec{u}_2\| = \sqrt{\frac{3}{2}} \Rightarrow \vec{x}_2 =$

$\|\vec{u}_3\| = \frac{2}{\sqrt{3}} \Rightarrow \vec{x}_3 =$

ortogonais $\perp \wedge \| \; \| \neq 1$ $\Longrightarrow$ ortonormados $\perp \wedge \| \; \| = 1$

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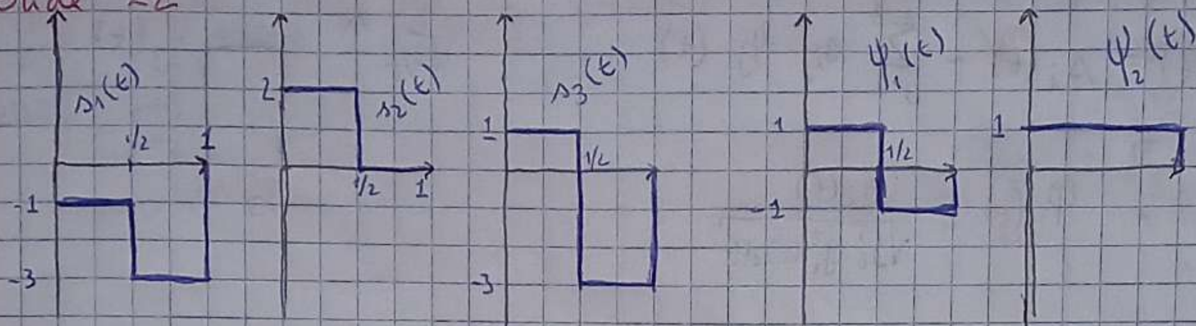

$$\int_0^T \psi_1(t) \cdot \psi_2(t)\, d(t) = \emptyset \ ?$$

$$\int_0^{T/2} 1\, dt - \int_{T/2}^{T} 1\, dt = [t]_0^{T/2} - [t]_{T/2}^{T}$$
$$= \frac{T}{2} - \left(T - \frac{T}{2}\right)$$
$$= \frac{T}{2} - \frac{T}{2}$$
$$= \emptyset$$

$s_1(t) = \psi_1(t) - 2\psi_2(t)$ $\quad s_1 = (1, -2)$

$s_2(t) = \psi_1(t) + \psi_2(t)$ $\quad s_2 = (1, 1)$

$s_3(t) = 2\psi_1(t) - \psi_2(t)$ $\quad s_3 = (2, -1)$

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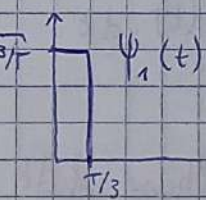
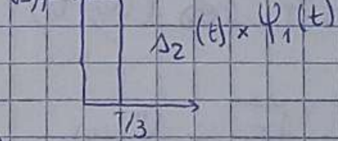
$$E_1 = \int_0^T s_1^2(t)\, dt = \int_0^{T/3} 1^2\, dt = [t]_0^{T/3} = \frac{T}{3}$$

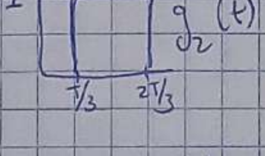
$$\psi_1(t) = \frac{s_1(t)}{\sqrt{E_1}} = \frac{s_1(t)}{\sqrt{T/3}}$$



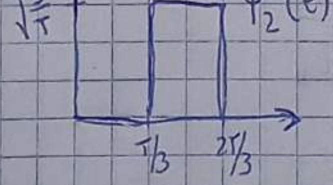
$$s_{2,1} = \int_0^T s_2(t) \times \psi_1(t) = \sqrt{\frac{T}{3}}$$



$$g_2 = s_2(t) - s_{2,1}\psi_1(t) =$$

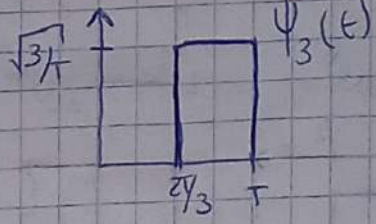


$$\psi_2(t) = \frac{g_2(t)}{\sqrt{\int_0^T g_2^2(t)\, dt}}$$

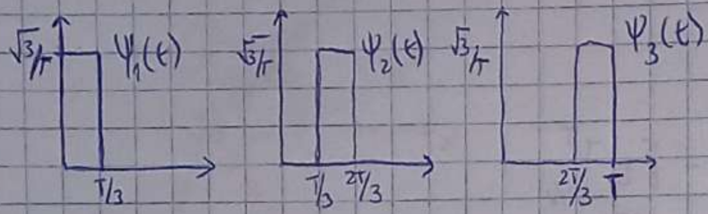


$\psi_3(t) = ?$

$$g_i(t) = s_i(t) - \sum_{j=1}^{i-1} s_{ij}\, \psi_j(t)$$



$$\psi_i(t) = \frac{g_i(t)}{\sqrt{\int_0^T g_i^2(t)\,dt}}$$



$s_1(t) = \sqrt{\frac{T}{3}} \times \psi_1(t)$

$s_2(t) = \sqrt{T/3} \times (\psi_1(t) + \psi_2(t))$

$s_3(t) = \sqrt{T/3} \times (\psi_2(t) + \psi_3(t))$

$s_4(t) = \sqrt{T/3} \times (\psi_1(t) + \psi_2(t) + \psi_3(t))$

Aula Teórica 5 (10/10/2023)

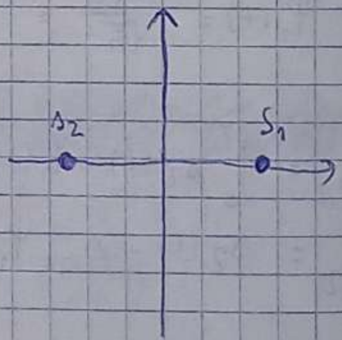
$s(t) = A\cos(w_c t + \phi)$

$E = \frac{A^2}{2} T$ (se for sinusóide) $\rightarrow A = \sqrt{\frac{2E}{T}}$

$E = \int_0^T s^2(t)\,dt$

Bynary Phase-Shift Keying (BPSK)

$\psi_1(t) = \sqrt{\frac{2}{T}} \cos(w_c t)$ (função base)

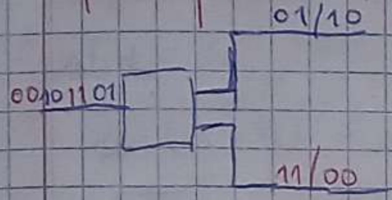


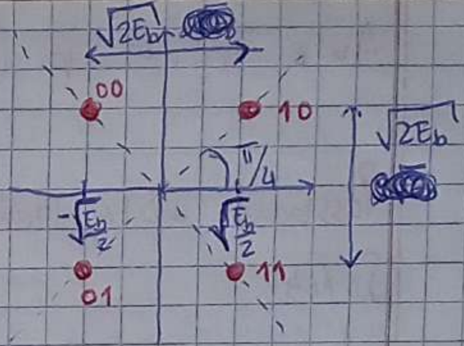
$$\begin{cases} s_1(t) = \sqrt{E_b} \times \psi_1(t) \\ s_2(t) = -\sqrt{E_b} \times \psi_1(t) \end{cases}$$

$P_e = Q\left(\frac{d}{\sqrt{2N_0}}\right)$, $d = 2\sqrt{E_b}$

$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

Quadriphase - Shift Keying (QPSK)





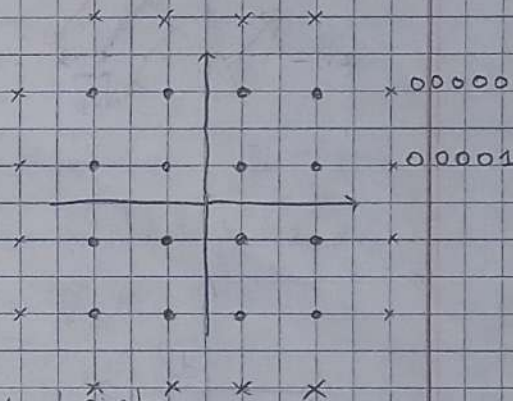
Quadrature - Amplitude Modulation (QAM)

Square: $M = 8, 16, 32, 64$

Cross: $M = 7, 15,$

$M = 4 \Rightarrow L = \sqrt{M} = \sqrt{4} = 2$

$2^m = 16 \Rightarrow 16\,QAM$



Exercício:

$M = 5 \Rightarrow ?$

$2^5 = 32$ pontos

$M - 1 = 4 \rightarrow$ square constellation with $2^4 = 16$ points

$2^{M-3} = 2^2 = 4$

Aula Teórica 6 (17/10/2023)

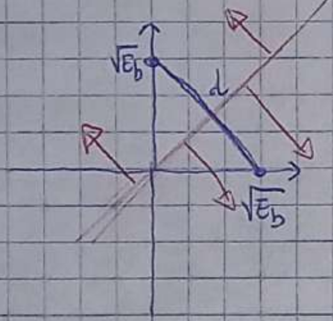
Passband Data Transmission

Slide 4

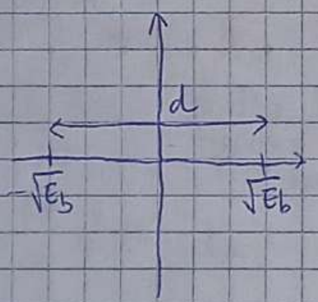
provar que $s_1(t)$ e $s_2(t)$ são ortogonais

$$\Rightarrow \int_0^{T_b} s_1(t) \times s_2(t)\, dt \quad \text{(produto interno)}$$

para cada modulação:
- moduladores
- probabilidade de erro
- largura de banda
- espetro



$d = \sqrt{2E_b} \rightarrow$ FSF



$d = 2\sqrt{E_b} \rightarrow$ PSK

para a mesma relação $\frac{E_b}{N_0}$, PSK tem uma menor probabilidade de erro $\Leftarrow$ uma d maior.

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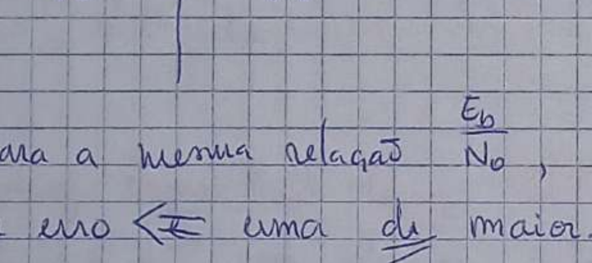

Slide 17 $\rightarrow$ BFSK (verde) é melhor nas questões de interferência intersimbólica, pq n̄ tem oscilações

MSK (minimum shift Keying

slide 21

$$\cos(a+b) = \cos(a)\cos(b) - \operatorname{sen}(a)\operatorname{sen}(b)$$

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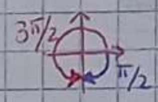
$\theta(0) = 0 \wedge \theta(T_b) = \frac{\pi}{2} \Rightarrow 1$

$\theta(0) = \pi \wedge \theta(T_b) = \frac{\pi}{2} \Rightarrow 0$

$\theta(0) = \pi \wedge \theta(T_b) = -\frac{\pi}{2} = \frac{3\pi}{2} \Rightarrow 1$

$\theta(0) = 0 \wedge \theta(T_b) = -\frac{\pi}{2} \Rightarrow 0$



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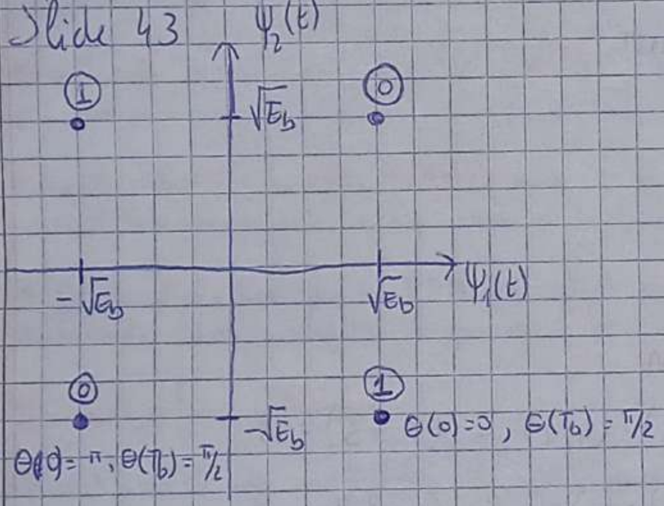

MSK tem a mesma eficiência em termos de probabilidade de erro BSK (?)

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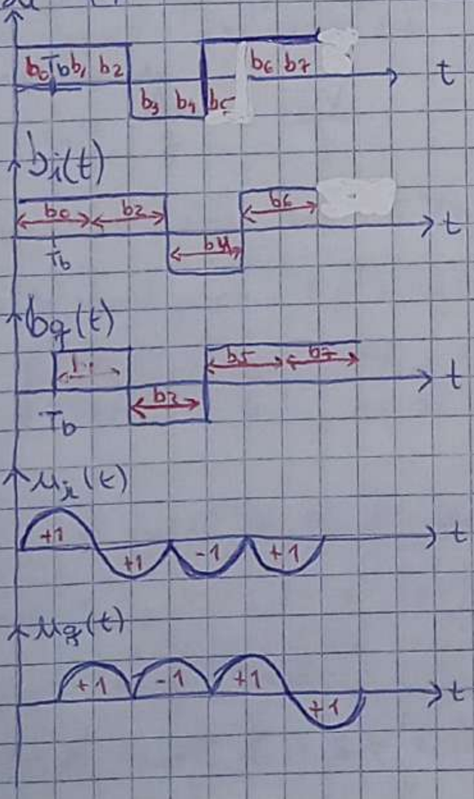

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$$\begin{cases} \varphi_1(t) = \sqrt{\frac{2}{T_b}} \cdot \cos\left(\frac{\pi t}{2T_b}\right) \cdot \cos(2\pi f_c t) \\ \varphi_2(t) = \sqrt{\frac{2}{T_b}} \cdot \operatorname{sen}\left(\frac{\pi t}{2T_b}\right) \cos(2\pi f_c t) \end{cases}$$

$$s(t) = s_1(t) \cdot \varphi_1(t) + s_2(t) \cdot \varphi_2(t)$$

$$\cos a \cdot \cos b = \frac{1}{2}\left[\cos(a-b) + \cos(a+b)\right]$$

$$\cos a + \cos b = 2\cos\left(\frac{a+b}{2}\right) \cdot \cos\left(\frac{a-b}{2}\right)$$

$$\cos a - \cos b = 2\operatorname{sen}\left(\frac{a+b}{2}\right) \cdot \operatorname{sen}\left(\frac{a-b}{2}\right)$$

$$f_1 + f_2 = \frac{\sqrt{2/T}}{2} \times \left(\cos\left(2\pi\left(f_c + \frac{1}{4T_b}\right)t\right) + \cos\left(2\pi\left(f_c - \frac{1}{4T_b}\right)t\right)\right)$$

$$= \frac{\sqrt{2/T}}{\cancel{2}} \times \cancel{2} \times \left(\cos\left(\frac{2\pi t \times (2f_c)}{\cancel{2}}\right) \cdot \cos\left(\frac{2\pi t \times \frac{1}{2T_b}}{2}\right)\right)$$

$$= \sqrt{2/T} \times \left(\cos(2\pi t f_c) \cdot \cos\left(\frac{\pi t}{2T_b}\right)\right)$$

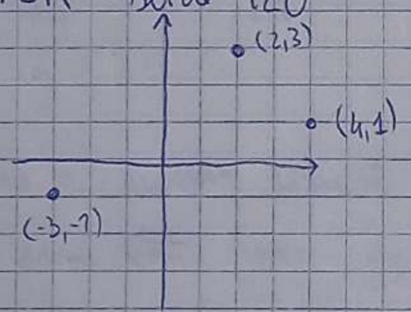
Gaussian-Filtered MSK (GMSK)

$$x(t) \rightarrow \boxed{\text{MSK}} \rightarrow \boxed{H(f)} \rightarrow \text{GMSK}$$

M-ARY FSK

Noncoherent Ortogonal Modulation

DPSK - slide 120



$(2,3) \cdot (4,1) = (8+3) = 11$

$(2,3) \cdot (-3,-1) = 6 - 3 = -9$